

Thrashing and Working Set Model

Memory Management – Virtual Memory

Prashant Jadav

DEPSTAR - CHARUSAT

Learning Objectives

By the end of this topic, students should be able to:

- Explain the concept of thrashing and its causes.
- Relate thrashing to the degree of multiprogramming and locality.
- Define the working set of a process.
- Calculate the working-set size for a given reference string.
- Use the working-set model to control thrashing.

Thrashing

Thrashing occurs when a process spends more time handling page faults and swapping pages than executing useful instructions.

Typical situation

The total number of frames required by active processes exceeds the available physical frames.

$$\sum_{i=1}^n WSS_i > m$$

where

- WSS_i = working-set size of process i
- m = total number of available physical frames

Effect

High page-fault rate $\rightarrow$ high disk I/O $\rightarrow$ low CPU utilization $\rightarrow$ poor system performance.

Why Does Thrashing Occur?

- ① A process needs more pages than the frames currently allocated to it.
- ② A page fault occurs frequently.
- ③ The OS brings the required page from secondary storage.
- ④ Another required page may be evicted immediately.
- ⑤ The cycle repeats continuously.

Important observation

Increasing the degree of multiprogramming does **not** always increase CPU utilization. Beyond a certain point, it can cause thrashing.

Too many active processes $\Rightarrow$ insufficient frames per process $\Rightarrow$ thrashing

Working Set Model

The **working set** of a process is the set of distinct pages referenced during the most recent Δ page references.

Definition

$$W(t, \Delta) = \{\text{pages referenced during the last } \Delta \text{ references at time } t\}$$

- Δ is called the **working-set window**.
- $WSS(t, \Delta) = |W(t, \Delta)|$ is the working-set size.
- The working set represents the process's current **locality**.

Key idea

A process should have enough frames to contain its current working set.

Working Set Example 1 – Basic

Reference string:

1, 2, 3, 1, 4, 2, 5, 1, 2, 3

Working-set window:

$\Delta = 4$			
t	Reference	Last 4 References	WSS
1	1	1	1
2	2	1,2	2
3	3	1,2,3	3
4	1	1,2,3,1	3
5	4	2,3,1,4	4
6	2	3,1,4,2	4
7	5	1,4,2,5	4
8	1	4,2,5,1	4
9	2	2,5,1,2	3
10	3	5,1,2,3	4

$$\max(WSS) = 4,$$

$$\text{Average WSS} = 3.2$$

Example 1 – Interpretation

$$WSS_{\max} = 4$$

Max. 4 frames are required if the process is to keep its entire current working set in memory.

Observation

The working-set size changes over time because the process moves between different localities.

$$W(7, 4) = \{1, 4, 2, 5\}$$

$$WSS(7, 4) = 4.$$

Later:

$$W(9, 4) = \{2, 5, 1\}$$

$$WSS(9, 4) = 3.$$

Example 2 – GATE-Type Calculation

Consider the page-reference string:

$c, c, d, b, c, e, c, e, a, d$

Working-set window:

$$\Delta = 4$$

Assume initially:

$$W(0, 4) = \{a, d, e\}.$$

Determine:

- 1 Total number of page faults.
- 2 Average number of frames required.

Example 2 – GATE-Type Calculation

Consider the page-reference string:

c, c, d, b, c, e, c, e, a, d

Working-set window:

$$\Delta = 4$$

Assume initially:

$$W(0, 4) = \{a, d, e\}.$$

Method

At each reference:

- 1 Consider the current reference and previous $\Delta - 1$ references.
- 2 Remove references outside the window.
- 3 Count only distinct pages.
- 4 Compare the current page with the resident working set.

Example 2 – Solution

t	Ref.	Working Set	WSS	Fault
1	c	{e,d,a,c}	4	Yes
2	c	{d,a,c}	3	No
3	d	{a,c,d}	3	No
4	b	{c,d,b}	3	Yes
5	c	{d,b,c}	3	No
6	e	{d,b,c,e}	4	Yes
7	c	{b,c,e}	3	No
8	e	{c,e}	2	No
9	a	{c,e,a}	3	Yes
10	d	{c,e,a,d}	4	Yes

Page faults = 5

$$\text{Average frames} = \frac{4 + 3 + 3 + 3 + 3 + 4 + 3 + 2 + 3 + 4}{10} = 3.2$$

Example 3 – Detecting Thrashing

Suppose a system has 10 physical frames and three active processes:

$$WSS_1 = 4, \quad WSS_2 = 5, \quad WSS_3 = 3$$

Total working-set demand:

$$D = \sum WSS_i = 4 + 5 + 3 = 12$$

Available frames: $m = 10$ Therefore:

$$D > m$$

Conclusion

The system cannot accommodate all 3 WS simultaneously. Thrashing is likely to occur.

OS action

Reduce the degree of multiprogramming.

Working Set Model and Thrashing Control

For n processes:

$$D = \sum_{i=1}^n WSS_i$$

Compare D with the total available frames m .

Condition	OS interpretation
$D < m$	Enough frames; processes can coexist
$D \approx m$	High frame pressure; monitor carefully
$D > m$	Thrashing likely; reduce multiprogramming

Main principle

The OS tries to ensure:

$$\sum_{i=1}^n WSS_i \leq m$$

Thrashing: Symptoms and Remedies

Symptoms

- Very high page-fault rate
- Excessive disk I/O
- Low CPU utilization
- Poor response time and throughput

Remedies

- Reduce degree of multiprogramming.
- Suspend one or more processes.
- Allocate sufficient frames to active processes.
- Use the working-set model to estimate frame demand.
- Page-Fault Frequency (PFF) can also be used as a control mechanism.

Exam / GATE Quick Questions

① If $\Delta = 5$, how many page references are considered when calculating the working set?

② If

$$WSS_1 = 5, WSS_2 = 4, WSS_3 = 3, \quad m = 10,$$

is thrashing likely?

③ Why can increasing the degree of multiprogramming reduce CPU utilization?

④ For the reference string

1, 2, 1, 3, 2, 4

and $\Delta = 3$, calculate the working set at $t = 6$.

Answers

- Q1: 5 references.
- Q2: Yes, because $5 + 4 + 3 = 12 > 10$.
- Q3: Processes receive fewer frames, increasing page faults and causing thrashing.
- Q4: $W(6, 3) = \{2, 4\}$, so $WSS = 2$.

Summary

- **Thrashing**: excessive paging with very little useful CPU work.
- Main cause: insufficient frames for the active processes.
- **Working Set** captures the pages currently used by a process.
- Working-set size:

$$WSS(t, \Delta) = |W(t, \Delta)|$$

- Total frame demand:

$$D = \sum WSS_i$$

- To avoid thrashing:

$$D \leq m$$

- If $D > m$, reduce the degree of multiprogramming.